

Associated Primes, Primary Decomposition, Support of a Module

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April 2025

1 Associated Primes

Understanding every topic should begin with examples, so let's start collecting them.

The theory of associated primes aims to extend the notion of unique factorization domains.

Let I be an ideal of R . By associated primes of I , we mean the associated primes of the module R/I instead of the module I .

This is because the associated primes of the ideal I are usually useless. For example, if R is a domain and I is any non-zero ideal, the associated primes of I is the singleton set containing the zero ideal.

Let's first show that if R is Noetherian, the set of associated primes of M is non-empty.

Consider the collection of all ideals of R that are annihilators of some $m \in M$. Let I be a maximal element of this collection. Such an I exists by the Noetherian hypothesis. Our goal will be to show that I is a prime ideal.

Let $xy \in I$ but assume $y \notin I$, since otherwise we're done. Our goal now is to show that $x \in I$.

Let $I = \text{Ann}(m)$ for some $m \in M$. The only hypothesis we have is that I is maximal, so we should use this hypothesis.

We have that $(xy)m = 0$ but $ym \neq 0$. Thus, $x \in \text{Ann}(ym)$. Also notice that I is contained in $\text{Ann}(ym)$ by the commutativity of R . Thus, if $x \notin I$, $\text{Ann}(ym)$ is an ideal that contains I and x , but it's not the unit ideal since $\text{Ann}(ym)$ is not the zero ideal. We have thus proven the following:

Lemma 1.1. Every module M over a Noetherian ring R satisfies $\text{Ass}(M)$ being non-empty.

Question: What is a ring that is not Noetherian and violates this hypothesis?

Let's first follow ATOCA Chapter 17.

Lemma 1.2. Let R be a ring, M be a module, and I be an ideal of R contained in $\text{Ann}(M)$. Then, there's a bijection between $\text{Ass}_R(M)$ and $\text{Ass}_{R/I}(M)$ given by induced bijection between the prime ideals given by the canonical surjection map.

Proof. Let P be a prime ideal of R such that there exists some $m \in M$ and $P = \text{Ann}(m)$. Then, I is contained in P .

Recall that since I is contained in $\text{Ann}(M)$, M is an R/I -module by simply extending the action. (This can most easily be seen using the ring endomorphism definition of a module.) This immediately implies that $r + I \in \text{Ann}(m)$ if and only if $r \in \text{Ann}(m)$, and thus we get the bijective correspondence we're looking for. $\square$

This lemma justifies letting $\text{Ann}(M) = 0$ without loss of generality when we are concerned with the associated primes of a module M .

Proposition 1.3. Let R be a ring, S be a multiplicative subset, M be an R -module, and P be a prime ideal of R .

Let $P \in \text{Ass}(M)$ such that $P \cap S = \emptyset$. Then, $S^{-1}P \in \text{Ass}(S^{-1}M)$.

Proof. Let $P \in \text{Ass}(M)$. Then, R/P injects into M . Since localization is exact, $S^{-1}(R/P) = S^{-1}R/S^{-1}P$ injects into $S^{-1}M$.

Recall that $S^{-1}P = P(S^{-1}R)$, which is a prime ideal of the localized ring.

Thus, $S^{-1}P$ is in $\text{Ass}(S^{-1}M)$. □

Proposition 1.4. Let R be a ring and M be an R -module. Assume either that R is Noetherian or that M is Noetherian. Then, $z.\text{div}(M)$ is the union of all associated primes of M .

Proof. Notice that the union of all associated primes is contained in $z.\text{div}(M)$ by definition of an associated prime. Let's now prove the reverse inclusion using both hypotheses.

Let $x \in z.\text{div}(M)$. Then, there exists $0 \neq m \in M$ such that $xm = 0$. Then, $x \in \text{Ann}(m)$.

Assuming R is Noetherian, $\text{Ann}(m)$ lies in an ideal P maximal among annihilators of nonzero elements, which is an associated prime of M .

Assume now that M is Noetherian. □

Corollary 1.4.1. Let R be a ring and

Proof. Notice that $0 \notin z.\text{div}(M)$ if and only if $M = 0$. Since every prime ideal P contains 0, $M = 0$ if and only if $\text{Ass}(M) = \emptyset$. □

Proof. □

Lemma 1.5. Let $M = M_1 \oplus M_2$. Then, $\text{Ass}(M) = \text{Ass}(M_1) \cup \text{Ass}(M_2)$.

Proof. Since M_1 and M_2 are submodules of M , we have that $\text{Ass}(M_1) \cup \text{Ass}(M_2)$ is contained in $\text{Ass}(M)$.

Proposition 17.6 in ATOCA produces the other containment. □

Corollary 1.5.1. Let $M = M_1 \oplus M_2 \oplus \cdots \oplus M_n$. Then, $\text{Ass}(M) = \text{Ass}(M_1) \cup \text{Ass}(M_2) \cup \cdots \cup \text{Ass}(M_n)$.

2 Primary Decomposition

The following definition is from ATOCA, and it's a pretty general definition. I will ease my way into it by using Gathmann's notes as well.

I also like the ATOCA definition better since it explains some of the asymmetry in the definition.

Definition 1. Let R be a ring and $Q \subsetneq M$ be R -modules. Q is called a **P -primary** module if $\text{Ass}(M/Q) = \{P\}$. Q is called **old-primary** if given $x \in R, m \in M$ either $m \in Q$ or $x \in \text{nil}(M/Q)$. Borchers calls these **primary submodules**.

Here's a definition from Borchers' lecture on the Lasker-Noether Theorem:

Definition 2. A module M is called **coprimary** if it has at most one associated prime.

Lemma 2.1. Let M be a finitely generated module over a Noetherian ring R . Let Q be a submodule of M . Q is P -primary if and only if M/Q is coprimary.

This is immediate, but showing the equivalence with old-primary submodules requires a tiny bit more work.

Lemma 2.2. Let M be a finitely generated module over a Noetherian ring R . Let Q be a submodule of M . Q is old-primary if and only if M/Q is coprimary.

Proof. Assume Q is old-primary. We'll just work with $N := M/Q$.

Let $0 \neq n \in M$. Let $P = \text{Ann}(n)$.

Notice that $\text{Ann}(N) \subseteq P$.

Also notice that $P \subseteq z.\text{div}(N) \subseteq \text{nil}(N) = \sqrt{\text{Ann}(N)}$. as Q is old-primary.

If P is prime, $\sqrt{\text{Ass}(N)} = P$.

Thus, there's at most one associated prime, which is $\sqrt{\text{Ann}(M)}$.

Conversely, assume $N := M/Q$ is coprimary.

Without loss of generality, assume $\text{Ann}(M) = 0$.

Let P be the only associated prime of N . Then, P contains $\text{Ann}(n)$ for every $n \in N$ since maximal elements of this form are associated primes.

We'll show that P is nilpotent. It suffices to show that every element $a \in P$ is nilpotent as P is finitely generated as R is Noetherian.

Assume there exists $a \in P$ is not nilpotent. Then, the localization M_a is not zero as

Then, the set of associated primes of M_a is not empty. Let T be an associated prime of M_a . Then, T is a prime ideal of R_a . The preimage of T under the canonical localization map

□

Theorem 2.3. Any finitely generated module M over a Noetherian ring is contained in the direct sum

$$\bigoplus_{P \in \text{Ass}(M)} M_P$$

where M_P are P -primary modules.

Notice that prime ideals are both primary and old-primary. Also notice that the notion of being primary depends on the submodule M .

Let's now show that this definition agrees with the primary ideal definition from Gathmann's commutative algebra notes.

Lemma 2.4. An ideal I in a ring R is old-primary if and only if for every $a, b \in R$ with $ab \in I$ we have $a \in I$ or $b \in I^n$, or equivalently, $b \in \sqrt{I}$.

Proof. Assume I is old-primary. Then, for every $x, y \in R$ such that $xy \in I$, either $x \in I$ and $y \in \text{nil}(R/I)$. Since $y \in \text{nil}(R/I)$ if and only if $y \in \sqrt{I}$, we're done. $\square$

Lemma 2.5 (Example 8.10 in Gathmann's CA). Let R be a PID. Then, an ideal I is old-primary if and only if $I = (p^n)$ for some prime $p \in R$.

Proof. Let's first prove the converse. Let $I = (p^n)$.

Let $x, y \in R$ such that $xy \in I$. Then, $xy = p^n r$ for some $r \in R$.

Consider the irreducible decompositions of x and y . If p divides y , we're done. Otherwise, p^n divides x and we're done again.

Now, assume $I = (x)$ is an old-primary ideal. Consider the irreducible decomposition of x . Clearly, if x has two distinct prime divisor, we can pick x and y to mention distinct prime divisors and this would imply that the primary condition doesn't hold. $\square$

Notice in fact that the argument above works for any principal ideal in a UFD. Also notice that if Q is an old-primary ideal, $P := \sqrt{Q}$ is a prime ideal. In this case, Q is sometimes called **P -old-primary**.

Let us now return to the general definition of ATOCA.

Lemma 2.6. Let R be a ring and $Q \subsetneq M$ be R -modules. Q is old-primary if and only if zero divisors on M/Q correspond to nilpotents on M/Q , i.e. $z \cdot \text{div}(M/Q) = \text{nil}(M/Q)$.

Proof. Assume Q is old-primary. Let $y \in R$ be a zero divisor on M/Q , i.e. $\square$

Thus, being an old-primary module is really a property of the way in which Q is embedded in M , i.e. it's a property of M/Q .

Lemma 2.7. Let R be a ring and $Q \subsetneq M$ be R -modules. If Q is old-primary, $\text{nil}(M/Q)$ is the smallest prime ideal containing $\text{Ann}(M/Q)$.

Proof. Let's first show that $\text{nil}(M/Q)$ is prime.

Let $x, y \in R$ such that $xy \in \text{nil}(M/Q)$. Assume $y \notin \text{nil}(M/Q)$.

Let Q be a prime ideal containing $\text{Ann}(M/Q)$. $\square$

Lemma 2.8 (ATOCA 18.4). Let R be a ring and N be a Noetherian R -module. Let $I = \text{nil}(N)$. Then, $I^n N = 0$ for some $n \geq 1$.

Proof. Let $J = \text{Ann}(N)$. Then, $I = \sqrt{J}$.

We know that R/J is Noetherian. Since R/J is Noetherian, $(I + J)/J$ is finitely generated. Since I is contained in $\sqrt{J}$ and is finitely generated, we have that there exists some $n \geq 1$ such that $I^n \subseteq J = \text{Ann}(N)$. In other words, $I^n N = 0$. $\square$

Proposition 2.9 (ATOCA 18.5). Let Q be a submodule of M . Assume Q is P -primary and M/Q is Noetherian.

Then, $P = \text{nil}(M/Q)$ and $P^n(M/Q) = 0$ for some $n \geq 1$.

Proof. $\square$

Proposition 2.10 (ATOCA 18.8). Let R be a ring and I be a maximal ideal of R . Let $Q \subsetneq M$ be R -modules.

Assume $\text{nil}(M/Q) = I$. Then, Q is old-primary.

Example 2.11 (Gathmann Example 8.13(a)). We show that primary ideals don't need to be powers of prime ideals in general.

Let $Q = (x^2, y)$ and $P = (x, y)$ in $\mathbb{R}[x]$.

Notice that $\sqrt{Q} = P$, which is a maximal ideal. Thus, Q is P -primary.

However, $P^2 = (x^2, xy, y^2)$ is strictly contained in Q .

Example 2.12 (Gathmann Example 8.13(b)). Let $R = \mathbb{R}[x, y, z](xy - z^2)$.

Let $P = (\bar{x}, \bar{z})$ be an ideal of R .

Notice that $P^2 = (\bar{x}^2, \bar{x}\bar{z}, \bar{z}^2)$ is not primary since $\bar{x}\bar{y} = \bar{z}^2$ is in P^2 but $\bar{x} \notin P^2$ and no power of $\bar{y}$ is in P^2 .

This example also highlights the asymmetry of the definition of an old-primary ideal, which is already apparent in the ATOCA definition.

Definition 3. Let I be an ideal in a ring R . A primary decomposition of I is a finite collection of old-primary ideals $Q_1, \dots, Q_n$ such that $I = Q_1 \cap Q_2 \cdots Q_n$.

Lemma 2.13. Let R be a ring and $I \subset Q$ be ideals of R . Q is old-primary in R if and only if Q/I is old-primary in R/I .

Proof. Recall that Q is old-primary if and only if zero-divisors in R/Q are nilpotent. Since R/Q is isomorphic to $(R/I)/(Q/I)$, we have the desired result. $\square$

Proposition 2.14. In a Noetherian ring, every ideal has a primary decomposition.

Proof. Assume by contradiction that there is an ideal without a primary decomposition.

Since R is Noetherian, there's an ideal I of R that is maximal among all ideals without primary decompositions.

Then, in $S = R/I$, the zero ideal is not a primary ideal. Then, there exists $a, b \in S$ such that $ab = 0$ but $a \neq 0$ and b is not nilpotent.

Since S is Noetherian, the chain of ideals $\text{Ann}(b) \subseteq \text{Ann}(b^2) \cdots$ becomes stationary at some $n \geq 1$.

Since (a) and (b^n) are non-zero ideals of S , they have a primary decomposition. Then, $(a) \cap (b^n)$ also has a primary decomposition.

Let $x \in (a) \cap (b^n)$. Then, $x = ac = db^n$ for some $c, d \in S$.

Then, $0 = cab = xb = db^{n+1}$, so $d \in \text{Ann}(b^{n+1}) = \text{Ann } b^n$, so $x = 0$.

We now have a contradiction since we obtained a primary decomposition of the zero ideal. $\square$

Example 2.15. Let $R = \mathbb{R}[x, y]/(x^2 + y^2 - 1)$ be the coordinate ring of the unit sphere over the reals. Let $X = V(x^2 + y^2 - 1)$.

Intuitively, $x^2 + y^2 - 1$ is irreducible as the circle would be a union of two lines otherwise, but it clearly is not.

Rigorously, $x^2 + y^2 - 1$ is irreducible as

We will show that R is not a UFD by showing that $\bar{x}$ is irreducible but not prime.

Notice that $V(\bar{x})$ consists of two points, the north and south poles of the sphere.

Thus, $\bar{x}$ is not zero since otherwise $V(\bar{x}) = X$.

Similarly, $\bar{y}$ is not a unit since otherwise $V(\bar{y}) = \emptyset$.

The statement that $\bar{x}$ is not prime simply corresponds to the geometric fact that $V(\bar{x})$ is reducible, as it's a union of two points.

Lemma 2.16. $\bar{x}$ is not prime.

Proof. We will show that $\bar{x}$ doesn't divide $1 \pm \bar{y}$. Since $\bar{x}^2 = (1 + \bar{y})(1 - \bar{y})$, this shows that $\bar{x}$ is not prime since $(1 + \bar{y})(1 - \bar{y})$ is in the ideal $(\bar{x})$ but neither factor is in $(\bar{x})$.

Assume $\bar{x}$ divides $1 + \bar{y}$.

Then, $1 + y = xg + (x^2 + y^2 - 1)h$ for some $g, h \in \mathbb{R}[x]$.

Plugging in $(x, y) = (0, 1)$ produces $2 = 0$, which is a contradiction. □

Lemma 2.17. $\bar{x}$ is irreducible.

Proof. □

Notice what the issue is here. $\bar{x}$ can't be written as a product of two functions, but the ideal $(\bar{x})$ is the intersection of the ideals $I(\bar{x}, \bar{y} - 1)$ and $I(\bar{x}, \bar{y} + 1)$ using the Nullstellensatz and that intersections of ideals correspond to unions of the varieties.